

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

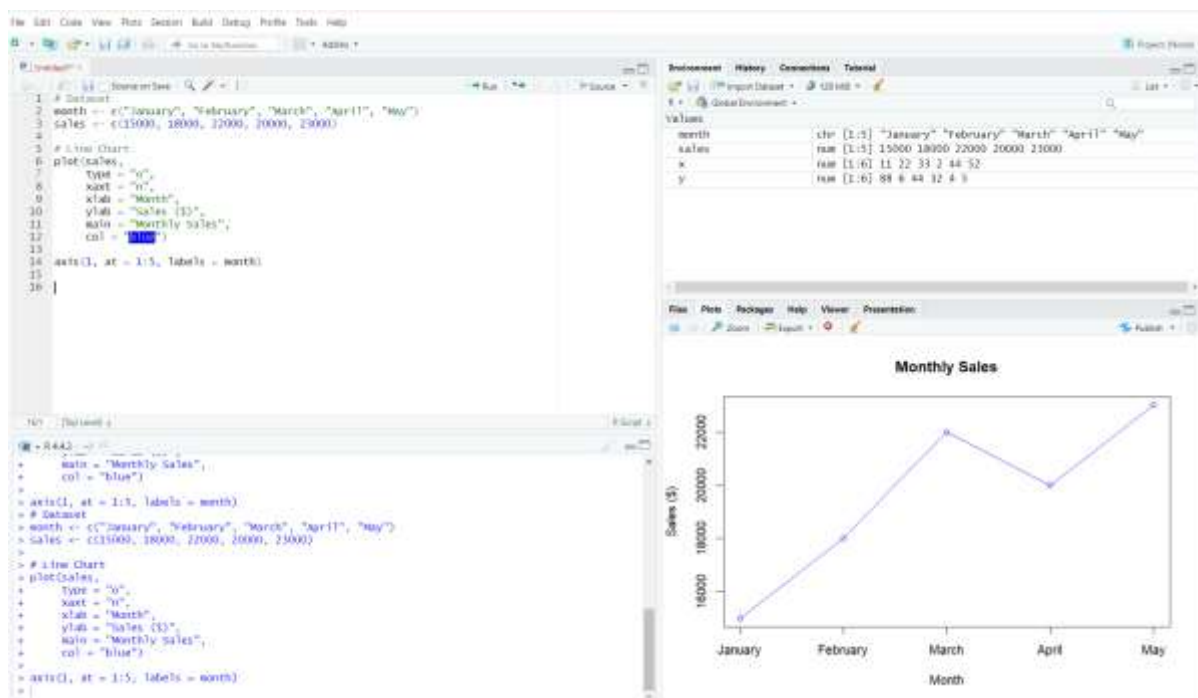
PROGRAM 1: Line Chart of Monthly Sales

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Line Chart
plot(sales,
     type = "o",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales",
     col = "blue")

axis(1, at = 1:5, labels = month)
```



Output 1: Monthly Sales Line Chart

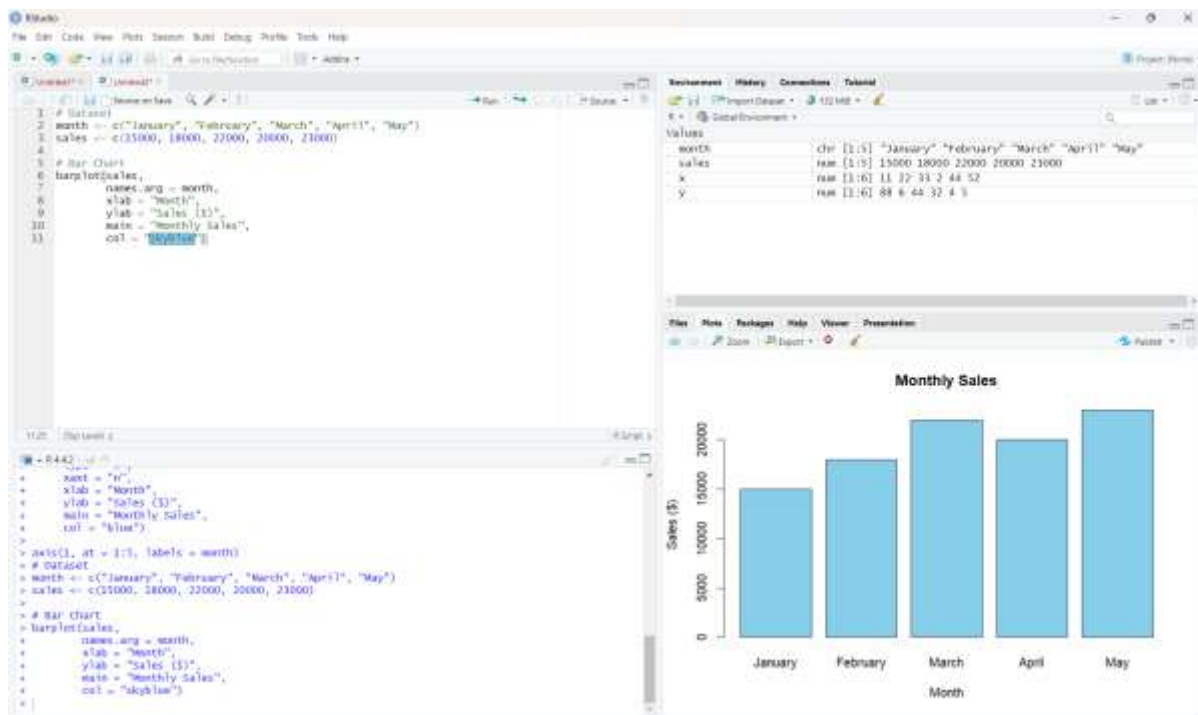
Observation: Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Sales

Generate a bar chart to display the sales values for the available months and label the chart elements.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Bar Chart
barplot(sales,
        names.arg = month,
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales",
        col = "skyblue")
```



Output 2: Monthly Sales Bar Chart

Observation: The bar chart makes it easy to compare the sales of each month. May has the highest sales and January has the lowest.

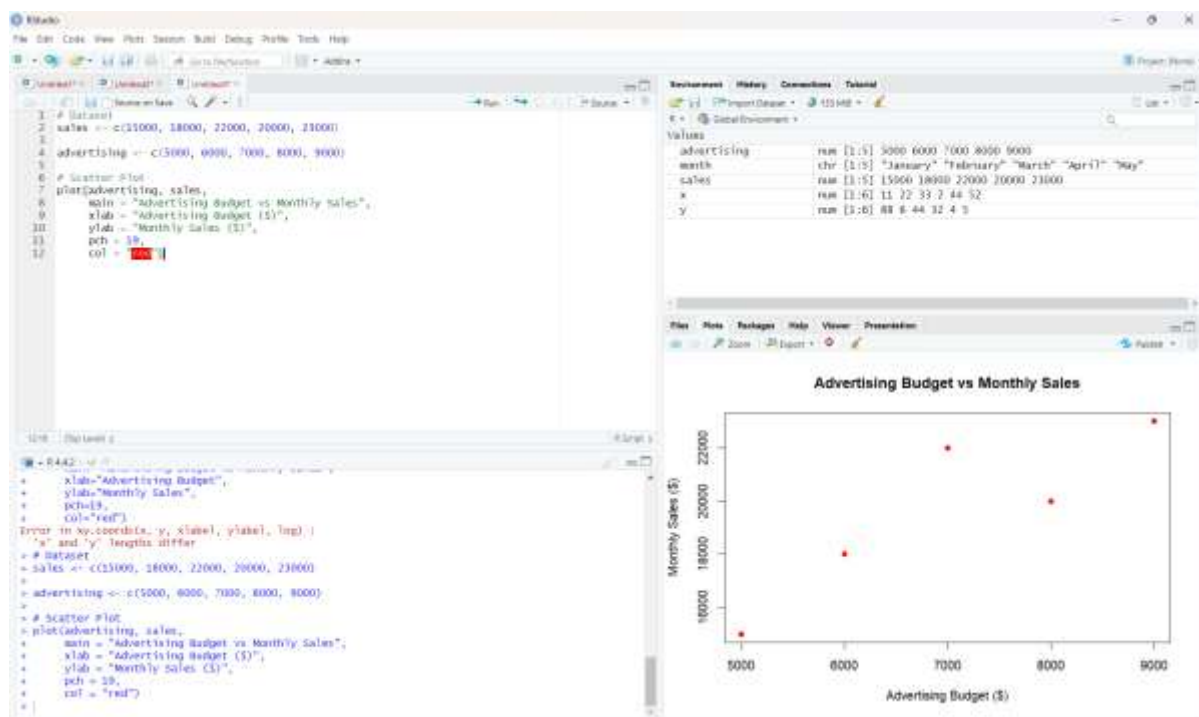
PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

```
# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)

advertising <- c(5000, 6000, 7000, 8000, 9000)

# Scatter Plot
plot(advertising, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "red")
```



Output 3: Advertising Budget vs Monthly Sales Scatter Plot

Insight: The plotted points show a generally positive relationship: as the advertising budget increases from \$5,000 to \$9,000, monthly sales also generally increase. This suggests that higher advertising spending is associated with higher sales for the values plotted.

Note: The visible assignment table contains only Month and Sales. The advertising values used in the submitted R program are the five values shown in the user's executed program: 5000, 6000, 7000, 8000, and 9000.

PROGRAM 4: Interactive Sales Dashboard

Build an interactive dashboard combining the line chart and bar chart using the Shiny package.

```
library(shiny)

month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput("chart",
                  "Select Chart:",
                  choices = c("Line Chart", "Bar Chart"))
    ),

    mainPanel(
      plotOutput("salesPlot")
    )
  )
)

server <- function(input, output) {

  output$salesPlot <- renderPlot({

    if (input$chart == "Line Chart") {

      plot(sales,
           type = "o",
           xaxt = "n",
           xlab = "Month",
           ylab = "Sales ($)",
           main = "Monthly Sales")

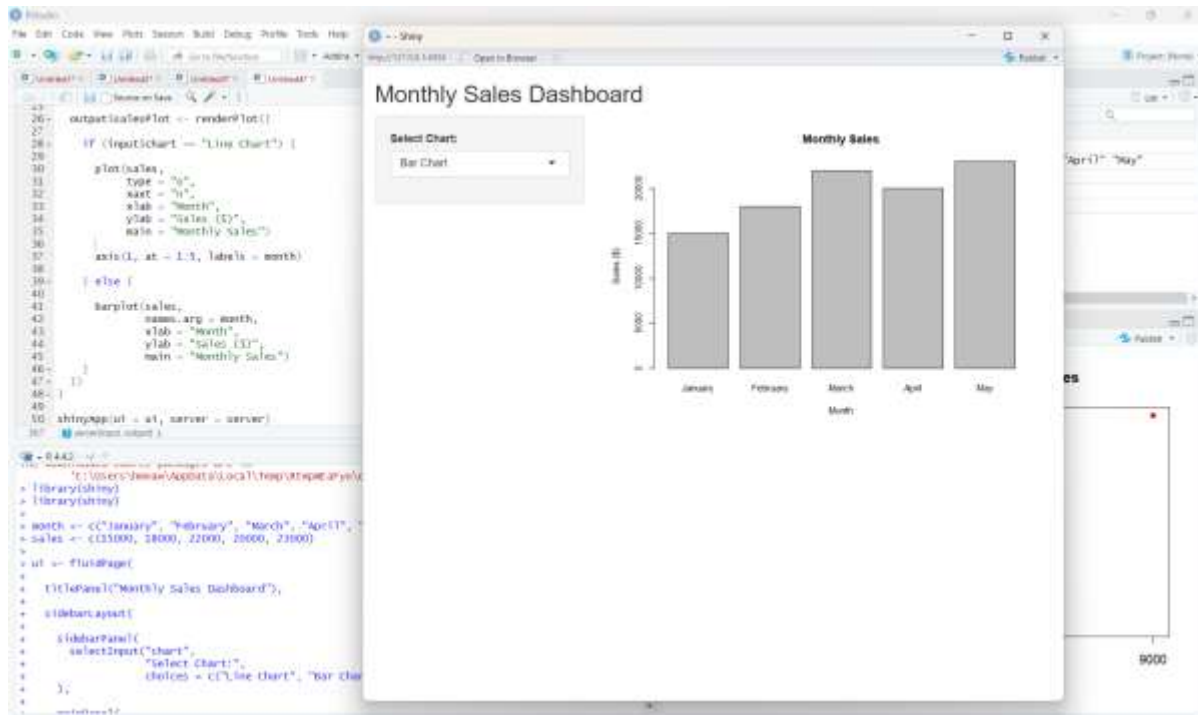
      axis(1, at = 1:5, labels = month)

    } else {

      barplot(sales,
              names.arg = month,
              xlab = "Month",
              ylab = "Sales ($)",
              main = "Monthly Sales")

    }
  })
}
```

```
shinyApp(ui = ui, server = server)
```



Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard provides a chart-selection control. Selecting Line Chart displays the monthly trend, while selecting Bar Chart displays the monthly sales comparison.

Procedure

1. Create the required month and sales vectors in R.
2. Use `plot()` to create the line chart with appropriate axis labels and title.
3. Use `barplot()` to compare the sales values for the available months.
4. Create advertising-budget and sales vectors of equal length and use `plot()` for the scatter plot.
5. Install and load the Shiny package, then create the UI and server components for the interactive dashboard.
6. Run the Shiny application and switch between the Line Chart and Bar Chart using the selection box.

Result

The monthly sales data was successfully visualized using line, bar, and scatter charts. An interactive Shiny dashboard was also created to switch between the line chart and bar chart.

Key R Functions Used

Function	Purpose
<code>plot()</code>	Creates line and scatter plots.
<code>axis()</code>	Adds custom axis labels.
<code>barplot()</code>	Creates bar charts.
<code>fluidPage()</code>	Creates the Shiny page layout.
<code>selectInput()</code>	Provides the chart selection control.
<code>renderPlot()</code>	Generates plots dynamically in Shiny.
<code>shinyApp()</code>	Runs the interactive Shiny application.